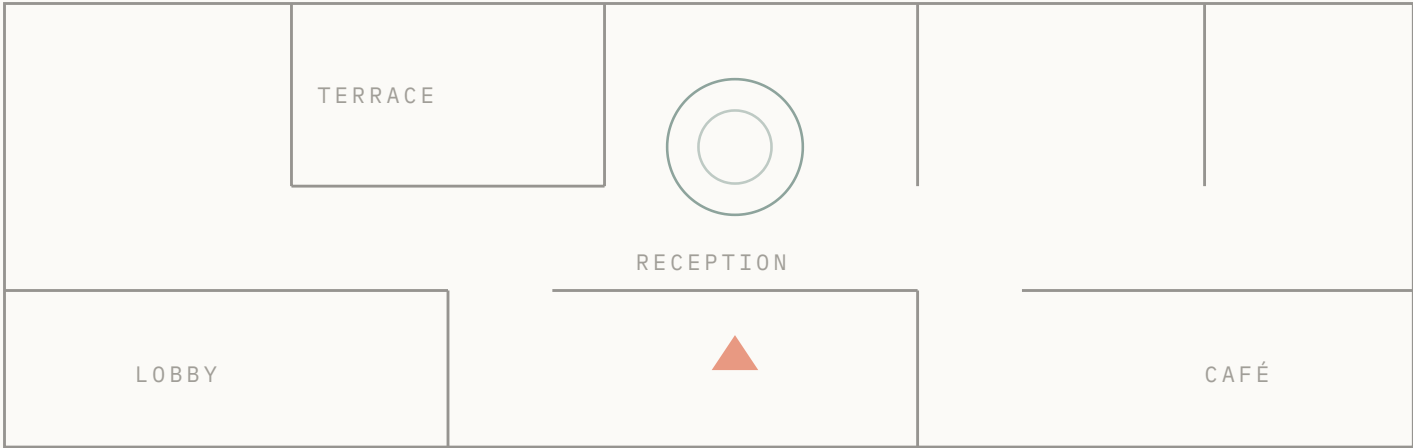


MÉYRAKI — SPATIAL INTELLIGENCE

Meyraki Insight

Build Plan & Recommendations

From floorplan to revenue: an AI agent pipeline that reads interior spaces, simulates guest flow, proposes better layouts, and delivers client-ready insight reports — for hospitality first, MENA first.



CLIENT Méyraki Studio	DOCUMENT Build Plan, Recommendations & Milestones	SHEET MI-001 · Rev A
PREPARED FOR Israa Sammoura · Jad El Khatib	SCOPE MVP — 8 weeks to pilot-ready product	STATUS Approved to start

What we found, what we build

Every Meyraki document — the notes, six decks, developer docs, mockups and samples — was read and merged into one source of truth.

Conflicts between timelines were resolved and recorded. In parallel, five research passes verified ~45 GitHub repositories, current AI-model pricing, and the competitive landscape across the USA, Europe and MENA.

The strategic finding: no product anywhere closes the loop *footfall* → *layout redesign* → *ROI forecast* → *client-ready report* for hospitality interiors. Analytics vendors stop at dashboards; AI-layout vendors have no behavioral data and don't serve hospitality. In MENA, no funded startup occupies this space at all.

We build a web SaaS: upload a floorplan and optional footfall data, choose a business objective, and a pipeline of seven specialized AI agents produces labeled zones, flow heatmaps, 2–3 optimized layout scenarios, a moodboard, and business metrics — assembled into a branded PDF your client can act on.

Everything the AI claims is validated: typed contracts between agents, a constraint solver guaranteeing layout feasibility, and a final QA agent that blocks inconsistent results. Assumptions ride with every number.

8 wks

to pilot-ready MVP

<3 min

full analysis, upload to report

<\$1.50

AI cost per analysis, enforced

02 · THE PRODUCT LOOP |

STEP	EXPERIENCE
Upload	Floorplan (PNG/JPG/PDF) + optional footfall CSV, or demo presets. Bad inputs are rejected politely at the door.
Objectives	Maximize guest flow · seating efficiency · revenue per sqm · ambiance — plus a free-text brief ("more sunlight, coworking feel").
Analysis	The agent pipeline runs; progress streams live, step by step, like a drawing register.
Insight	Heatmap over the plan, bottleneck pins, A/B layout scenarios with predicted effects, moodboard with palette and materials.
Report	Branded PDF in English or Arabic — the deliverable your client pays for.

The stack, decided

LAYER	DECISION	WHY
Frontend	Next.js (React + TypeScript), Tailwind, Konva canvas	Web SaaS with heavy canvas, reports and SEO — React owns this ecosystem. Flutter Web (old docs) is weak exactly there. Arabic/RTL from day one.
Backend	FastAPI (Python 3.12) + PostgreSQL + S3-compatible storage	The original docs were right — kept. Python is where every AI SDK and spatial library lives; Pydantic contracts double as agent-output validation.
AI engine	LLM-vision agent pipeline; OpenCV as deterministic tools inside it	Replaces the 2024 OpenCV+scikit-learn plan: frontier models read floorplans semantically with zero training data. Classical CV stays for masks, geometry, rendering — testable and free.
Models	Claude Sonnet 5 (vision) · Claude Opus 5 (layout reasoning) · Haiku 4.5 (cheap steps) · Gemini Flash Image (renders ~\$0.04)	Sonnet 5's high-resolution vision does pixel-accurate coordinates on architectural drawings; Opus 5 powers the step that is the product; Claude doesn't generate images, Gemini is best value for interior renders.
Provider strategy	One thin model abstraction (Pydantic AI); AWS Bedrock path for GCC	Provider swap is config, not code. Claude runs on Bedrock in UAE & Bahrain regions (verified) — the enterprise/ministry data-residency answer. EU in-region for the strongest compliance story.
Orchestration	Pydantic AI + Postgres-backed resumable job pipeline	Least framework that fits a deterministic 7-step graph with typed outputs. LangGraph is the documented alternative; AutoGen is in maintenance mode — avoided.
Reports	Branded HTML → Playwright PDF	One design system serves app and report; print-quality output; trivially embeds heatmaps and renders.

Adopt — commercially safe

pdfplumber (vector-PDF plans → exact wall geometry, MIT) · ezdxf (DWG/DXF path, MIT) · OR-Tools CP-SAT (layout constraints, Apache) · LayoutGPT pattern (LLM layout proposals, MIT) · JuPedSim (validated pedestrian simulation, Jülich) · node-vibrant (palettes, MIT) · Playwright (PDF, Apache) · warp & ThatOpen components (floor-map UI + browser CAD, MIT).

Blocked — looks tempting, legally unusable

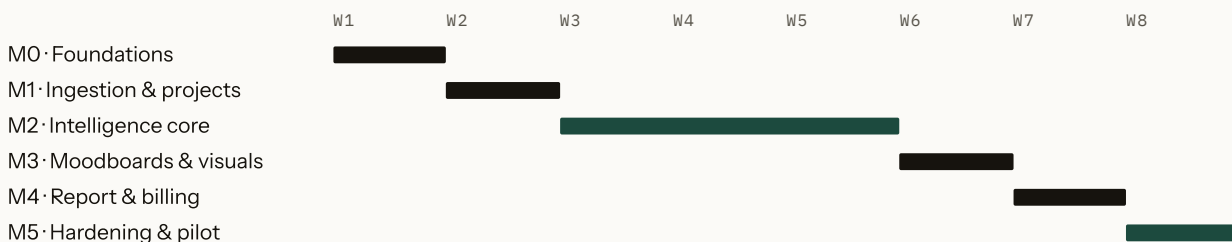
CubiCasa5k (non-commercial) · HouseDiffusion, House-GAN++, ATISS, DiffuScene (research-only riders) · Ultralytics YOLO & PyMuPDF (AGPL) · Graph2Plan (no license). Knowing this now avoids a legal rebuild later. Full audit: docs/04-REUSE-MAP.md.

Seven agents, step by step

Agents don't wander. A fixed, resumable graph runs them in order; every output is a validated, typed contract; a crashed job resumes from its last completed step. Steps 3–5 fan out in parallel (marked in green). Deterministic code handles everything deterministic — geometry, heatmap rendering, metric math — so models are only used where judgment is required.

1	Intake Agent HAIKU 4.5 · VISION	Analyzes what we have: validates the plan, checks footfall schema, infers space type, rejects garbage with a human reason. The trust boundary.
2	Routing Agent HAIKU 4.5	Decides the track — data-driven (CSV present) or simulated — which specialists run, model tier and cost budget per step.
3	Zone Analyst SONNET 5 · HIGH-RES VISION	Reads the drawing: rooms, walls, doors → labeled zone polygons with adjacency and walkable mask. Geometry code validates every polygon.
4a	Flow Analyst JUPEDSIM + RENDERER	Joins footfall to zones, or simulates guests through the walkable mask. Outputs per-zone intensity, bottlenecks, dead zones, heatmap.
4b	Layout Optimizer OPUS 5 + OR-TOOLS CP-SAT	Proposes 2–3 concrete scenarios ("move reception to north wall") with predicted effects; the constraint solver guarantees feasibility — clearances, capacity, egress.
4c	Moodboard Designer SONNET 5 + GEMINI IMAGE	Plans palette, materials, furniture era and lighting from the brief; renders 3–4 interior previews; extracts swatches.
5	Business Analyst DETERMINISTIC + SONNET 5	Computes ROI per sqm, flow efficiency, payback per scenario — math in code, model only explains. Assumptions always attached.
6	Report Writer SONNET 5 · EN / AR	Client-facing narrative in brand voice; deterministic HTML → PDF assembly with the client's identity.
7	QA Verifier SONNET 5 · SKEPTIC PROMPT	Cross-checks everything: referenced zones exist, numbers are consistent, no recommendation contradicts another. Fails → targeted re-run. Nothing unverified reaches a client.

The eight-week line



Green bars carry the highest technical risk and the most value; M5 extends through week 8 into pilot.

M0 Foundations

WEEK 1

- Monorepo: Next.js app · FastAPI api · shared contracts package (one schema source: Pydantic → generated TypeScript)
- Postgres, object storage, managed auth (org/multi-tenant from day one), CI: lint · typecheck · tests
- Design tokens from the new identity; EN/AR i18n scaffold, RTL verified on the first screen

EXIT Deployed skeleton — login, empty dashboard, health checks green.

M1 Ingestion & projects

WEEK 2

- Projects CRUD (client, logo, space type) · floorplan upload with validation · footfall CSV with inline schema errors · demo presets
- Job pipeline skeleton: Postgres queue, resumable typed steps, live progress stream, per-step audit rows

EXIT Upload → stub pipeline runs → progress streams live in the UI.

M2 The intelligence core

WEEKS 3–5

- All seven agents implemented in order (Intake → QA), typed contracts between every step
- Golden set first: ~20 hand-labeled floorplans; CI scores zone accuracy on every change
- Flow simulation (JuPedSim) + brand heatmap renderer · LLM-proposes/solver-guarantees layout scenarios

EXIT Real floorplan in → verified zones, heatmap, scenarios, metrics out.

M3 Moodboards & the visual wow

WEEK 6

- Moodboard Designer: style plan → generated interior renders → palette extraction → composed board
- Insight screen finalized: heatmap overlay, zone toggles, bottleneck pins, A/B scenario tabs

EXIT The four core screens exist for real — Upload → Objectives → Insight → Design.

M4 Report & the business loop

WEEK 7

- Branded report template (EN/AR) → print-quality PDF · report versioning
- Billing: subscriptions for studios, one-off per-report checkout

EXIT Upload → analyze → moodboard → client-ready branded PDF → paid.

M5 Hardening & pilot

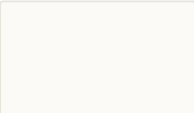
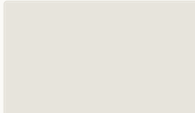
WEEK 8 →

- Failure drills: killed workers resume, poison inputs rejected, provider outage fails over, cost ceilings hold
- Security pass: signed URLs, org isolation, rate limits · 3–5 real pilot projects, prompts tuned on the golden set

EXIT Pilot-ready product + demo matching the StartupDen narrative.

The drawing sheet, *not* the dashboard

Modern, minimalist, high-class — as directed. The identity comes from Meyraki's own material: the architectural drawing. Gallery-white surfaces, ink linework, hairline rules, measured annotations. One deep accent; heat reserved for data. No gradients, no glow, no shadows.

					
PAPER #FBFAF7	INK #16130E	GRAPHITE #57534A	HAIRLINE #E7E4DC	VIRIDIAN #1C4A3E	THERMAL #DA4B22

HEATMAP RAMP — COOL → HOT · DATA ONLY, NEVER DECORATION

TYPOGRAPHY

Instrument Serif —
display & key numbers

Instrument Sans — interface & body

IBM PLEX MONO — ANNOTATIONS · METRICS ·
ZONE-04

Arabic: IBM Plex Sans Arabic for UI; display face
under evaluation. RTL mirrored layouts from day
one.

RULES OF THE SHEET

- The floorplan is always the hero; UI recedes around it
- One accent per view — viridian is brand, thermal is information
- Hairlines at 1px, radius 2px; depth from contrast, not shadows
- White space is the luxury cue; density is never the default
- Signature device: the dimension-line annotation (as on this page)
- Every AI number carries its assumptions, in mono, next to the number

How "no bugs" is enforced, not promised

GATE	MECHANISM
Typed contracts	Every agent output validated against a schema before the next step runs; malformed output is retried with the validation error. One schema source for API, pipeline and UI.
Golden set in CI	~20 hand-labeled floorplans; zone accuracy scored on every change — model or prompt drift fails the build before it reaches a client.
Solver guarantees	Layout scenarios must satisfy hard constraints (clearances, capacity, egress) in OR-Tools before they are shown. The model proposes; the solver proves.
QA Verifier	An independent skeptic agent cross-checks all outputs for consistency; failures trigger targeted re-runs, then human review. Inconsistent results are never delivered.
Resumable jobs	Steps are idempotent and persisted; crashes resume from the last completed step. Full audit trail: every prompt, output, cost and duration per job.
Honest numbers	Every metric ships with its assumptions; no black-box figures in front of a client or investor.

RISK	MITIGATION
Zone detection accuracy on messy scans	Golden-set benchmark from week 3; vector-PDF path bypasses vision entirely; GPL fallback model identified if LLM vision underperforms.
AI cost creep	Routing Agent enforces a per-analysis budget (< \$1.50); model tiers matched to step value.
Enterprise data residency (GCC)	Bedrock UAE/Bahrain path verified for Claude; EU in-region as the strongest compliance story. Same prompts, different endpoint.
Predicted-uplift credibility	Simulation-based forecasts labeled as such; pilot projects create the first real before/after evidence.

Milestone 0 starts now: monorepo scaffold, contracts package, design tokens from this document, CI, and the EN/AR shell. First demo target: the sample Cleo floorplan and footfall data from the source bundle, end to end.